Solving multi-step inequalities

1 Solve the following inequalities:

a $6x - 54 \geq 30$

b $21 + 7x \geq 21$

c $64 > 8x + 24$

d $3x + 27 > 3$

e $8 - x < 6$

f $-9 - x > 4$

g $4 \leq 1 - x$

h $4(x + 3) \leq 12$

i $-9(x + 6) \geq -27$

j $\dfrac{-x}{-2} > 6$

k $\dfrac{x}{2} - 1 \leq 7$

l $-3 \leq 1 + \dfrac{x}{6}$

m $\dfrac{x + 2}{3} \geq 2$

n $5 > \dfrac{x - 9}{2}$

o $\dfrac{x + 5}{-4} > 4$

p $7 - \dfrac{1}{7}x > 9$

q $-12 > 4(4x + 5)$

r $4(4 - 3x) \leq 28$

s $9(x - 7) + 5 \leq 5$

t $\dfrac{3x + 8}{5} > 4$

u $3\left(\dfrac{x}{5} - 6\right) \leq 9$

v $18 < 9\left(\dfrac{x}{5} - 6\right)$

w $2\left(8 - \dfrac{x}{3}\right) \geq 14$

x $7 < \dfrac{x - 2}{5} + 2$

2 Solve the following inequalities:

a $7x + 4 > x + 16$

b $\dfrac{x}{4} + \dfrac{x}{3} \geq -7$

c $2x + 6 > 30 - x$

d $\dfrac{9x}{5} + \dfrac{7x}{6} > 5$

e $\dfrac{2}{5}x + \dfrac{1}{4} < \dfrac{4}{5}x - \dfrac{1}{4}$

3 Consider the inequality $21 + 7x > 21$:

a Solve the inequality.

b State whether the following values of x satisfy the inequality $21 + 7x > 21$:

i $x = 0$ **ii** $x = 1$ **iii** $x = 14$ **iv** $x = 7$

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4 Consider the inequality $6(x+2) \leq 48$:

a Solve the inequality.

b State whether the following values of x satisfy the inequality $6(x+2) \leq 48$:

i $x = -6$ **ii** $x = 6$ **iii** $x = 0$ **iv** $x = 12$

5 Consider the inequality $3 - 3x < 2x - 4$:

a Solve the inequality.

b State whether the following values of x satisfy the inequality $3 - 3x < 2x - 4$:

i $x = 1$ **ii** $x = \frac{7}{5}$ **iii** $x = \frac{12}{5}$ **iv** $x = 2$

6 For each of the following inequalities:

i Solve the inequality.

ii Plot the solutions to the inequality on a number line.

a $2x - 2 < 6$

b $3 < 4x - 1$

c $-4x + 9 > 9$

d $3(x-1) < 9$

e $-7x - 5 \leq 2$

f $8 < 2(x+7)$

g $20 - 12x < 32$

h $\frac{x-7}{6} > -1$

i $1 \leq \frac{x-5}{3}$

j $2 - \frac{x}{3} \geq 0$

k $\frac{-8-3x}{2} \leq 5$

l $\frac{3x}{4} + 3 \geq -3$

m $67 - 9x \leq 7x + 19$

7 Isabelle tried to solve the following inequality but made a mistake in her work.

$$\begin{aligned} &\text{Step 0:} \frac{x}{4} + 2 < -3 \\ &\text{Step 1:} \quad \frac{x}{4} < -1 \\ &\text{Step 2:} \quad x < -4 \end{aligned}$$

Determine which step is incorrect and explain the error.

8 Construct and solve an inequality for the following situation: "The sum of 2 groups of x and 1 is at least 7."

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9 Consider the following situation: "2 less than 4 groups of p is no more than 18".

a Construct and solve the inequality.

b Find the largest value of p that satisfies this condition.

10 Sophia has a budget for school stationary of $\$32$, but has already spent $\$15.25$ on books and folders. Write an inequality that shows how much Sophia can spend on other stationary, where p represent the amount she can spend, and solve for p.

11 Xavier is not very good at saving his pocket money. Looking at the receipts for all of his purchases from the last week, Xavier can see that he has spent $\$33.40$. He really wants to go to the movies with his friends on the weekend but needs to have $\$32.00$.
Write an inequality that shows how much money Xavier must have had in his account at the beginning of the week to afford his weekend plans, where A represents the amount of money in Xavier's bank account at the beginning of the week and solve for A.

12 To get a grade of C, John must obtain a total score of at least 300 over his four exams. So far he has taken the first three exams and gotten scores of 63, 73, and 98.
If x represents what he must score on the last exam to get a C or better, write an inequality and solve for x.

13 At a sport clubhouse the coach wants to rope off a rectangular area that is adjacent to the building. He uses the length of the building as one side of the area, which measures 26 m. He has at most 42 m of rope available to use.
If the width of the roped area is W, form an inequality and solve for the range of possible widths.

14 Lachlan is planning on going on vacation. He has saved $\$2118.40$, and spends $\$488.30$ on his airplane ticket.

a Let x represent the amount of money Lachlan spends on the rest of his vacation. Write an inequality to represent the situation, and then solve for x.

b Find the highest amount that Lachlan could spend on the rest of his holiday.

15 Jimmy spends $\$120$ to produce some rockmelons. He is able to sell each one for $\$9$.

a If n represents the number of rockmelons he must sell to make a profit, write an inequality and solve for n.

b Find the least number of rockmelons that Jimmy must sell to make a profit.

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16 Skye was given $\$72$ for a birthday present. This present, along with earnings from a Saturday job, is being set aside for a mountain bike. The job pays $\$5.00$ per hour, and the bike costs $\$379$.

a Set up and solve an inequality to find the minimum number of hours that Skye needs to work to be able to buy the mountain bike, where h represents the number of hours worked. Round your answer to one decimal place.

b If Skye can only work her job for a whole number of hours, find the minimum number of hours she must work to afford her bike.

17 James wants to order some books from an online bookstore. Each book costs $\$13.30$ and shipping for the entire order is a flat rate of $\$29.05$. James can spend no more than $\$74.00$.

a If B represents the possible number of books that James can buy, write an inequality and solve for B. Round your answer to one decimal place, if necessary.

b Find the maximum number of books that James can afford to buy.

18 The table shows the times of six swimmers in a $1500\ \text{m}$ race. Only those who achieve a time of 22 minutes or below qualify for the final.
Use the inequality $t \leq 22$, where t is the time of each swimmer, to find all of the swimmers who qualify for the final.

Swimmer	Time (minutes)
James	21
Harry	24
Lachlan	20
Adam	27
Glen	31
Kenneth	23

19 Neville is saving up to buy an LED TV that is selling for $\$950$. He has $\$650$ in his bank account and expects a nice sum of money for his birthday next month.

a If the amount he is to receive for his birthday is represented by x, write an inequality that models the situation where he is able to afford the LED TV.

b How much money would he have in total if his parents were to give him $\$310$ for his birthday?

c Would he have enough to buy the plasma TV if his parents were to give him $\$310$ for his birthday?

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20 A particular bus can carry a maximum weight of $9500\ \text{lb}$ on board. If the average adult passenger weighs $150\ \text{lb}$ and the driver weighs $160\ \text{lb}$, then the number of passengers on board the bus at any one time must satisfy $150a + 160 \leq 9500$, where a represents the number of adult passengers.

- **a** There are 58 adults already on board and 8 more at the bus stop waiting to get on. Find the combined weight of everyone on board if the driver lets all of the new passengers on.
- **b** Can the driver safely let all of the new passengers on board?

Additional questions

21 Explain the differences between the following inequalities:

$0 \leq x < 4$ $\qquad$ $3 \leq x \leq 7$

22 Describe the range of values that satisfy each inequality:

- **a** $x \geq 29$ $\qquad$ **b** $x < -29$

23 When breeding certain types of fish it is recommended that the number of female fish is more than double the number of male fish.

- **a** Write the inequality for the recommended relationship. Let f be the number of female fish and m be the number of male fish.
- **b** State whether the following combinations align with the recommendation.
  - **i** $f = 10, m = 8$ $\qquad$ **ii** $f = 23, m = 9$
  - **iii** $f = 13, m = 10$ $\qquad$ **iv** $f = 7, m = 4$

24 James is saving up to buy a laptop that is selling for $\$550$. He has $\$410$ in his bank account and expects a nice sum of money for his birthday next month.

- **a** Write the inequality that models the situation in which James can afford the laptop. Let x represents the amount he is to receive for his birthday.
- **b** Plot the solution to the inequality on a numberline.

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25 Ryan wants to save up enough money so that he can buy a new sports equipment set, which costs $\$40.00$. Ryan has $\$22.10$ that he saved from his birthday. In order to make more money, he plans to wash neighbors' windows for $\$2$ per window.

a Let w be the number of windows that Ryan washes. Solve for w, correct to two decimal places.

b State whether the following statements are correct. Explain your thinking.

i Ryan must wash more than 9 windows to be able to afford the equipment.

ii Ryan must wash at least 8 windows to be able to afford the equipment.

iii If Ryan washed 8 windows, and 95% of another window, he could afford the equipment.

iv The number of windows Ryan must wash to be able to afford the equipment must be greater than or equal to 9.

26 Percy tried to plot the solution to the inequality $4x + 28 \geq -8$ on a numberline, however his answer is incorrect.



Identify the error(s) and explain how to rectify them.

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